

PUJITHA PADMANABHA

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Summary

ECE final-year student with a strong testing and validation mindset built across hardware, embedded, and industrial domains. Grounded in networking fundamentals and passionate about fault analysis and system reliability, looking to apply that to systems test engineering.

Experience

Indian Institute of Science

Jun 2026 – Present

Project Associate I, Analog and RF Systems Lab

Bengaluru

- Working on integration of embedded systems and communication protocols RF/signal systems at ARSL

Frauscher Sensor Technology

Nov 2025 – May 2026

Application Engineering Intern

Bengaluru

- Conducted hardware testing and signal analysis for railway sensor systems, including system configuration and performance validation
- Analyzed PCB circuits and proposed optimized testing sequences to improve diagnostic efficiency

IEEE Circuits and Systems Society

Oct 2025 – Dec 2025

Intern

Bengaluru

- Implemented iterative ChaCha20 cryptographic core (8/12/20 rounds) in Xilinx Vivado on FPGA achieving up to 4.27 Gbps throughput

ABB India Limited

Jul 2025 – Oct 2025

Student Intern

Bengaluru

- Worked on industrial drive systems (VFDs); analyzed faults and supported troubleshooting of power electronics systems
- Developed a digital fault tracking system to improve maintenance workflow and reporting efficiency

Robomanthan Pvt Ltd

Aug 2024 – Sept 2024

Robotics Intern

Bengaluru

- Built ESP32 and Arduino-based robotic systems including gesture-controlled lighting, Bluetooth car bots, and OpenCV color tracking; developed a Python simulation of a Roomba-style robot

Samsung Innovation Campus

Oct 2023 – Dec 2023

Student Trainee

Bengaluru

- Performed Python-based data analysis on Kaggle datasets to detect fraudulent job postings

Projects

- ASIC-Oriented RTL Implementation of ASCON-128** – Designed Verilog implementation comparing serial and unrolled architectures with mixed-cycle PPA optimization and FPGA validation
- MSHR-Based Non-Blocking Cache Verification** – Designed a 2-way set associative non-blocking cache with 4-entry coalescing MSHRs and verified using a self-checking Verilog testbench
- Music Reactive Lights using IoT** – Implemented Arduino-based lighting system reacting to audio signals with customizable effects
- Network Traffic Anomaly Detection** – Applied MATLAB-based ML techniques to detect DDoS, MITM, and SQL injection patterns with 90% accuracy

Education

BNM Institute of Technology

June 2026

B.E. Electronics and Communication

GPA: 9.83/10

Technical Skills

- HDL & Programming:** Verilog, Python, MATLAB, C
- EDA & Hardware Tools:** Cadence Genus/Virtuoso, Xilinx Vivado, STM32CubeIDE, Arduino IDE, PSpice
- Core Areas:** Digital Communication, DSP, Wireless Networks, RTL Design, Computer Architecture, Verilog Verification, CMOS Fundamentals
- Certifications:** NPTEL (Physical C-Based VLSI Design, Design for Testability), Cisco Networking, Samsung Innovation Campus (Python)

Publications

- Co-author, "Quantum Dots in Modern Medicine: Advancing Imaging, Diagnostics, and Drug Delivery", IEEE IITCEE 2025
- Co-author, "ASIC-Oriented RTL Implementation of ASCON-128", published in IJSEI